

THE FRULLANI INTEGRAL

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When I was first learning Real Analysis through Spivak's book, it seemed as if the subject wasn't "taking off" in the sense that the theorems felt disjoint, with each seemingly obvious result demanding a difficult proof.

Although I am not aware of any applications of the Frullani Integral in other branches of Mathematics, proving it (with help from a Polish postgraduate) was when I first felt I was doing serious Mathematics.

I have, therefore, decided to type up my original proof of this result.

We assume that f is continuous.

The Frullani Integral

Suppose that $\frac{f(x)}{x}$ is integrable on every interval $[a, b]$ for $0 < a < b$, and that $\lim_{x \rightarrow 0} f(x) = A$ and $\lim_{x \rightarrow \infty} f(x) = B$. Then, for all $\alpha, \beta > 0$, we have the identity

$$\int_0^\infty \frac{f(\alpha x) - f(\beta x)}{x} dx = (A - B) \log \frac{\beta}{\alpha}.$$

Proof: Consider the integral

$$\int_\varepsilon^N \frac{f(\alpha x) - f(\beta x)}{x} dx = \int_\varepsilon^N \frac{f(\alpha x)}{x} dx - \int_\varepsilon^N \frac{f(\beta x)}{x} dx.$$

We employ the substitution $u = \alpha x$ and $v = \beta x$ to the above equation, yielding (note $du = \alpha dx$ and $dv = \beta dx$)

$$\int_\varepsilon^N \frac{f(\alpha x) - f(\beta x)}{x} dx = \int_{\alpha\varepsilon}^{\alpha N} \frac{f(u)}{u} du - \int_{\beta\varepsilon}^{\beta N} \frac{f(v)}{v} dv.$$

Since v is a dummy variable, we can write

$$\int_\varepsilon^N \frac{f(\alpha x) - f(\beta x)}{x} dx = \int_{\alpha\varepsilon}^{\alpha N} \frac{f(u)}{u} du - \int_{\beta\varepsilon}^{\beta N} \frac{f(u)}{u} du.$$

Assume, without loss of generality, that $\alpha < \beta$. Then we may write

$$\begin{aligned}
\int_{\varepsilon}^N \frac{f(\alpha x) - f(\beta x)}{x} dx &= \int_{\alpha\varepsilon}^{\beta\varepsilon} \frac{f(u)}{u} du + \int_{\beta\varepsilon}^{\alpha N} \frac{f(u)}{u} du - \int_{\beta\varepsilon}^{\beta N} \frac{f(u)}{u} du \\
&= \int_{\alpha\varepsilon}^{\beta\varepsilon} \frac{f(u)}{u} du - \int_{\alpha N}^{\beta N} \frac{f(u)}{u} du.
\end{aligned}$$

The First Integral Mean Value Theorem implies that there exist numbers $k_1 \in [\alpha\varepsilon, \beta\varepsilon]$ and $k_2 \in [\alpha N, \beta N]$ such that

$$\begin{aligned}
\int_{\varepsilon}^N \frac{f(\alpha x) - f(\beta x)}{x} dx &= f(k_1) \int_{\alpha\varepsilon}^{\beta\varepsilon} \frac{1}{u} du - f(k_2) \int_{\alpha N}^{\beta N} \frac{1}{u} du \\
&= (f(k_1) - f(k_2)) \log \frac{\beta}{\alpha}.
\end{aligned}$$

Sending $\varepsilon \rightarrow 0$, $f(k_1) \rightarrow A$ and sending $N \rightarrow \infty$, $f(k_2) \rightarrow B$, so it follows that

$$\int_0^{\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx = (A - B) \log \frac{\beta}{\alpha},$$

as required. ■

We can deduce a corollary:

Corollary.

Let $\int_a^{\infty} f(x) dx$ converge for all $a > 0$, and suppose $\lim_{x \rightarrow 0} f(x) = A$. Then

$$\int_0^{\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx = A \ln \frac{\beta}{\alpha}.$$

Proof: To prove this, we consider the integral $\int_{\varepsilon}^{\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx$. Applying the same substitutions and assuming that $\beta > \alpha$, we obtain that

$$\int_{\varepsilon}^{\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx = \int_{\alpha\varepsilon}^{\beta\varepsilon} \frac{f(u)}{u} du.$$

The First Integral Mean Value Theorem then implies there exists $k \in [\alpha\varepsilon, \beta\varepsilon]$ such that

$$\int_{\varepsilon}^{\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx = f(k) \ln \frac{\beta}{\alpha}$$

Sending $\varepsilon \rightarrow 0$, $f(k) \rightarrow A$ which demonstrates

$$\int_0^{\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx = A \ln \frac{\beta}{\alpha},$$

as desired. ■